

STEVENSVILLE, MD, USA

WMO: 720384

Lat: 38.967N

Lon: 76.333W

Elev: 17

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 00124

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.9	20.9	-0.4	5.4	20.1	3.3	6.5	22.7	26.1	29.5	24.1	32.0	10.9	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	91.0	75.6	89.5	75.1	86.5	73.6	78.3	86.4	77.2	84.7	76.0	83.5	7.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	139.3	82.0	74.9	130.8	81.0	73.2	123.5	79.7	41.9	86.1	40.5	84.2	39.4	83.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.1	16.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	WB	11.5	80.5	4.1	2.1	8.6	82.0	6.2	83.3	3.9	84.4	0.9	85.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	35.6	38.0	44.9	56.5	66.2	74.6	78.7	76.6	71.2	60.4	48.2	40.9		
	DBStd	16.75	8.44	8.84	8.81	8.01	7.04	5.18	4.37	4.32	5.85	7.76	7.29	7.64		
	HDD50	1460	451	348	211	27	0	0	0	0	0	8	117	298		
	HDD65	4092	912	755	625	274	73	3	0	0	12	186	506	746		
	CDD50	4292	4	13	51	223	501	738	891	824	637	331	62	17		
	CDD65	1448	0	0	0	20	109	291	426	359	198	44	1	0		
	CDH74	11026	0	1	1	130	668	2152	3970	2743	1169	191	1	0		
	CDH80	3255	0	0	0	20	155	625	1366	771	284	34	0	0		
	WSAvg	6.7	7.6	7.5	7.5	8.0	6.5	6.0	5.5	5.0	5.7	6.7	7.4	7.2		
	PrecAvg	46.0	3.5	2.6	4.3	3.6	3.7	4.2	4.4	4.5	4.3	3.9	3.1	3.7		
(15) Precipitation	PrecMax	65.9	6.9	7.7	9.3	6.6	8.2	12.7	8.5	8.1	12.3	8.2	6.5	8.2		
	PrecMin	35.3	1.4	0.3	0.0	1.5	0.7	2.2	1.6	1.2	0.6	0.1	0.6	1.4		
	PrecStd	6.9	1.3	1.7	1.9	1.2	2.0	2.4	2.0	1.9	2.7	2.4	1.9	1.7		
	PrecStd	6.9	1.3	1.7	1.9	1.2	2.0	2.4	2.0	1.9	2.7	2.4	1.9	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	68.2	72.5	82.1	88.5	93.0	95.4	93.1	90.3	83.7	70.4	64.1		
		MCWB	55.3	58.8	59.1	67.6	74.3	75.4	78.2	75.6	74.4	72.8	62.1	60.9		
	2%	DB	55.4	61.2	67.9	78.5	84.1	89.8	91.3	90.0	86.4	79.1	65.7	60.9		
		MCWB	51.2	54.9	57.0	63.8	72.2	74.3	76.2	74.7	73.1	69.8	59.9	58.0		
	5%	DB	52.1	55.0	63.3	72.9	81.0	86.3	89.8	87.7	82.4	75.1	62.7	56.7		
		MCWB	47.7	48.9	54.5	62.2	69.7	73.1	76.1	73.9	71.7	69.7	56.7	53.8		
	10%	DB	46.5	51.6	59.0	69.9	77.4	82.5	87.6	84.0	80.8	72.3	59.3	52.3		
		MCWB	42.3	45.6	51.8	61.5	67.1	71.4	74.4	72.9	70.4	66.4	53.4	49.0		
	0.4%	WB	57.7	61.4	62.4	71.0	76.1	78.0	81.0	79.6	77.8	75.3	64.9	63.2		
		MCDB	61.0	65.7	66.9	78.1	85.6	87.8	89.9	88.1	85.0	80.1	68.3	64.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.8	56.3	59.4	67.7	73.4	76.3	79.0	77.9	75.4	72.5	61.4	58.9		
		MCDB	55.9	61.1	64.6	73.9	81.1	85.1	87.5	84.8	82.3	76.8	63.7	60.5		
	5%	WB	47.6	50.3	56.4	64.7	71.5	75.2	77.7	76.2	74.1	70.2	58.4	54.6		
		MCDB	51.3	54.2	62.2	71.2	78.6	83.3	85.5	82.9	80.1	74.3	61.8	56.8		
	10%	WB	43.8	45.9	53.0	62.1	69.8	74.0	76.6	74.9	72.8	67.3	55.1	49.9		
		MCDB	46.8	49.5	57.7	68.1	75.2	81.0	84.1	81.7	78.4	71.7	58.4	51.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	13.1	14.7	16.7	15.7	14.8	14.3	13.7	13.8	13.7	13.0	11.4		
		MCDBR	19.4	20.6	21.1	21.6	18.5	17.2	16.6	16.1	16.0	16.0	16.7	16.0		
		MCWBR	16.3	15.3	14.3	11.5	9.2	7.2	6.3	7.2	7.3	8.8	12.3	15.3		
	5% WB	MCDBR	18.5	19.9	19.9	20.0	17.0	16.0	14.7	14.0	14.1	14.0	14.8	15.7		
		MCWBR	16.9	17.0	15.4	12.5	8.7	7.4	6.5	6.8	7.4	9.2	12.9	15.7		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.320	0.361	0.398	0.443	0.494	0.518	0.511	0.422	0.362	0.326	0.311		
	taud		2.479	2.445	2.385	2.301	2.209	2.111	2.060	2.064	2.288	2.459	2.516	2.524		
	Ebn at Noon		274	287	284	279	267	253	245	243	260	268	267	266		
	Edh at Noon		25	30	35	40	45	50	52	51	38	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		663	932	1238	1581	1781	1968	1937	1720	1400	1043	754	582		
	RadStd		53	103	98	121	211	150	93	99	154	112	64	55		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+238	+140

Nomenclature: See separate page